COMP7103 Assignment 1

Data Mining

Name: _____ University No.: _____

Due 6 November 2026, 11:59 pm (Hong Kong)

Norms used throughout are the standard L_p norms. For a vector $V = (v_1, \ldots, v_d)$,

$$\|V\|_1 = \sum_{i=1}^{d} |v_i|, \qquad \|V\|_2 = \sqrt{\sum_{i=1}^{d} v_i^2}, \qquad \|V\|_\infty = \max_i |v_i|.$$

The absolute values are required for $\|\cdot\|_1$ and $\|\cdot\|_\infty$; otherwise these would not be norms. Cosine similarity and information gain use $\log_2$. Entropies and gains below are reported to 6 decimal places. A gain is compared with the pre-pruning threshold 0.15 before that rounding.

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1 Question 1: Similarity measures

Let the term order be (borrow, film, good, great, know, movie, part, think). The five term-frequency vectors are

$$\begin{aligned} R_1 &= (0, 0, 2, 2, 1, 5, 0, 2), \\ R_2 &= (0, 5, 0, 1, 3, 4, 2, 1), \\ R_3 &= (3, 0, 2, 0, 0, 10, 2, 2), \\ R_4 &= (0, 3, 0, 0, 1, 4, 0, 0), \\ R_5 &= (0, 1, 2, 2, 0, 4, 0, 0). \end{aligned}$$

Cosine similarity is

$$\cos(A, B) = \frac{A \cdot B}{\|A\|_2 \, \|B\|_2}.$$

1(a) Most similar reviews under term frequency

Squared lengths, obtained by summing the squared components:

$$\|R_1\|_2^2 = 38, \quad \|R_2\|_2^2 = 56, \quad \|R_3\|_2^2 = 121,$$
$$\|R_4\|_2^2 = 26, \quad \|R_5\|_2^2 = 25.$$

So $\|R_3\|_2 = 11$ and $\|R_5\|_2 = 5$. The ten pairwise dot products and cosine similarities are:

Pair	Dot product	$\|R_a\|_2^2$	$\|R_b\|_2^2$	Cosine
R_1, R_5	28	38	25	**0.908440**
R_2, R_4	34	56	26	0.891042
R_1, R_3	58	38	121	0.855349
R_3, R_5	44	121	25	0.800000
R_4, R_5	19	26	25	0.745241
R_3, R_4	40	121	26	0.713150
R_1, R_4	21	38	26	0.668099
R_2, R_5	23	56	25	0.614701
R_1, R_2	27	38	56	0.585299
R_2, R_3	46	56	121	0.558819

The largest value is $\cos(R_1, R_5)$. Its dot product expands as

$$R_1 \cdot R_5 = 2 \cdot 2 + 2 \cdot 2 + 5 \cdot 4 = 4 + 4 + 20 = 28,$$

and

$$\cos(R_1, R_5) = \frac{28}{\sqrt{38} \cdot 5} = \frac{28}{5\sqrt{38}} = 0.908440.$$

The next largest value is $\cos(R_2, R_4) = 0.891042$, so the maximum is unique.

Answer. The most similar pair under term frequency is reviews 1 and 5, with cosine similarity 0.908440.

1(b) TF–IDF and the most similar pair

There are $|D| = 5$ reviews. Term frequency is the raw count. Document frequencies and $\text{idf}(t) = \log_2(|D|/|D_t|)$ are:

Term	$\|D_t\|$	idf	value
borrow	1	$\log_2 5$	2.321928
film, good, great, know, think	3	$\log_2(5/3)$	0.736966
part	2	$\log_2(5/2)$	1.321928
movie	5	$\log_2 1$	0

Movie occurs in every review, so its idf is $\log_2 1 = 0$ and it drops out of every TF–IDF vector. Multiplying each raw count by the idf of its term gives

$$R_1 = (0,\ 0,\ 1.473931,\ 1.473931,\ 0.736966,\ 0,\ 0,\ 1.473931),$$
$$R_2 = (0,\ 3.684828,\ 0,\ 0.736966,\ 2.210897,\ 0,\ 2.643856,\ 0.736966),$$
$$R_3 = (6.965784,\ 0,\ 1.473931,\ 0,\ 0,\ 0,\ 2.643856,\ 1.473931),$$
$$R_4 = (0,\ 2.210897,\ 0,\ 0,\ 0.736966,\ 0,\ 0,\ 0),$$
$$R_5 = (0,\ 0.736966,\ 1.473931,\ 1.473931,\ 0,\ 0,\ 0,\ 0).$$

(Each displayed component is tf × idf rounded to 6 decimal places. The cosines below use the unrounded products.)

Pair	Dot product	Cosine
R_2, R_4	9.776129	**0.814237**
R_1, R_5	4.344946	0.739600
R_2, R_5	3.801828	0.333776
R_4, R_5	1.629355	0.316228
R_1, R_2	3.801828	0.277719
R_1, R_3	4.344946	0.211353
R_2, R_3	8.076212	0.202620
R_3, R_5	2.172473	0.127007
R_1, R_4	0.543118	0.087706
R_3, R_4	0	0.000000

For the winning pair, write $\alpha = \log_2(5/3)$. The only nonzero contributions to $R_2 \cdot R_4$ are film and know:

$$R_2 \cdot R_4 = (5\alpha)(3\alpha) + (3\alpha)(1\alpha) = 18\alpha^2 = 9.776129.$$

The Euclidean norms are $\|R_2\|_2 = 5.151916$ and $\|R_4\|_2 = 2.330490$, so

$$\cos(R_2, R_4) = \frac{9.776129}{5.151916 \times 2.330490} = 0.814237.$$

The runner-up is $\cos(R_1, R_5) = 0.739600$. Reweighting by idf changes the most similar pair: "movie" no longer contributes, and reviews 2 and 4 share the comparatively rare term "film".

Answer. The most similar pair under TF–IDF is reviews 2 and 4, with cosine similarity 0.814237.

2 Question 2: Metric axioms

A function d is a metric when it satisfies all three of the following, for every data object p, q, r.

- **Positivity.** $d(p, q) \geq 0$, and $d(p, q) = 0$ if and only if $p = q$. Objects are identified by their position and orientation, so $p = q$ means $P_p = P_q$ and $\Theta_p = \Theta_q$.
- **Symmetry.** $d(p, q) = d(q, p)$.
- **Triangle inequality.** $d(p, r) \leq d(p, q) + d(q, r)$.

Positions below are in millimetres. Orientation components are integers in $[0, 360)$. The difference $\Theta_p - \Theta_q$ is ordinary vector subtraction, as specified by the L_p norm of that difference.

2(a) $d_1(p, q) = \min(\|P_p - P_q\|_1, \|\Theta_p - \Theta_q\|_1)$

Positivity fails. Take

$$p: \ P = (0, 0, 0), \ \Theta = (0, 0, 0), \qquad q: \ P = (0, 0, 0), \ \Theta = (10, 0, 0).$$

Then $p \neq q$, but $\|P_p - P_q\|_1 = 0$ and $\|\Theta_p - \Theta_q\|_1 = 10$, so $d_1(p, q) = \min(0, 10) = 0$.

Symmetry holds. For any vector V, $\|V\|_1 = \|-V\|_1$. Therefore both arguments of the minimum are unchanged when p and q are swapped, and so is the minimum.

Triangle inequality fails. Take

$$\begin{aligned} a: \ & P = (0, 0, 0), \ \Theta = (0, 0, 0), \\ b: \ & P = (5, 0, 0), \ \Theta = (0, 0, 0), \\ c: \ & P = (5, 0, 0), \ \Theta = (5, 0, 0). \end{aligned}$$

Then

$$d_1(a,b) = \min(5,0) = 0,$$
$$d_1(b,c) = \min(0,5) = 0,$$
$$d_1(a,c) = \min(5,5) = 5.$$

Since $5 \not\leq 0+0$, the triangle inequality fails.

2(b) $d_2(p,q) = \|P_p - P_q\|_2 + \|\Theta_p - \Theta_q\|_\infty$

Write $d_P(p,q) = \|P_p - P_q\|_2$ and $d_\Theta(p,q) = \|\Theta_p - \Theta_q\|_\infty$. Euclidean distance and L_∞ distance are both metrics, and $d_2 = d_P + d_\Theta$.

Positivity holds. A sum of nonnegative numbers is nonnegative. The sum is zero only when both terms are zero, hence only when $P_p = P_q$ and $\Theta_p = \Theta_q$.

Symmetry holds. Both d_P and d_Θ are symmetric, so their sum is symmetric.

Triangle inequality holds. For any a, b, c,

$$d_P(a,c) \leq d_P(a,b) + d_P(b,c), \qquad d_\Theta(a,c) \leq d_\Theta(a,b) + d_\Theta(b,c).$$

Adding these two inequalities gives $d_2(a,c) \leq d_2(a,b) + d_2(b,c)$.

Thus d_2 satisfies all three axioms.

2(c) $d_3(p,q) = (\|P_p - P_q\|_2)^2 + \|\Theta_p - \Theta_q\|_1$

Positivity holds. Both terms are nonnegative, so $d_3(p,q) \geq 0$. The square of an L_2 norm is zero exactly when the vector is zero, and an L_1 norm is zero exactly when its vector is zero. Hence $d_3(p,q) = 0$ exactly when the position vectors agree and the orientation vectors agree. Also $d_3(p,p) = 0$.

Symmetry holds. Replacing a difference by its negative does not change either norm, and squaring does not change that fact.

Triangle inequality fails. Use three positions on a line and identical orientations, so the orientation term is zero:

$$a:\ P = (0,0,0),\ \Theta = (0,0,0),$$
$$b:\ P = (1,0,0),\ \Theta = (0,0,0),$$
$$c:\ P = (2,0,0),\ \Theta = (0,0,0).$$

Then $d_3(a,b) = 1^2 = 1$, $d_3(b,c) = 1$, and $d_3(a,c) = 2^2 = 4$. Since $4 \not\leq 1+1$, the triangle inequality fails. Setting the orientations equal isolates the squared Euclidean distance, which is what breaks the axiom; the added orientation term does not repair it.

2(d) $d_4(p,q) = \|P_p\|_2 - \|P_q\|_2 + \|\Theta_p\|_2 - \|\Theta_q\|_2$

Define $f(x) = \|P_x\|_2 + \|\Theta_x\|_2$. Then $d_4(p,q) = f(p) - f(q)$.

Positivity fails. Take

$$p:\ P = (0,0,0),\ \Theta = (0,0,0), \qquad q:\ P = (3,4,0),\ \Theta = (0,0,0).$$

Then $f(p) = 0$, $f(q) = \|(3,4,0)\|_2 = 5$, and $d_4(p,q) = -5 < 0$.

Symmetry fails. For the same pair, $d_4(q,p) = 5 \neq -5 = d_4(p,q)$.

Triangle inequality holds. For every triple p, q, r,

$$\begin{aligned} d_4(p,q) + d_4(q,r) &= \big(f(p) - f(q)\big) + \big(f(q) - f(r)\big) \\ &= f(p) - f(r) \\ &= d_4(p,r). \end{aligned}$$

The two sides are equal, so $d_4(p,r) \le d_4(p,q) + d_4(q,r)$ holds for every triple, including triples that produce a negative value.

3 Question 3: Classification

Entropy of a set D with class proportions p_c is $H(D) = -\sum_c p_c \log_2 p_c$, and a binary split into L and R has information gain

$$\text{Gain}(D) = H(D) - \frac{|L|}{|D|} H(L) - \frac{|R|}{|D|} H(R).$$

A node is split only when it contains at least 8 training instances and its best binary split has gain at least 0.15. Otherwise it becomes a leaf labelled with its majority class. Every leaf below has a unique majority, so no tie-break is required.

Smoking is nominal, so a binary split is a partition of its values, not a numeric threshold. Age is numeric. Each candidate threshold is the midpoint between two successive distinct ages present in the node; every instance with a given age then falls entirely on one side.

The tree

Smoking $\in \{1, 3\}$?
yes, and Age ≤ 62.5: predict **HC** (8 instances; best gain $0.143156 < 0.15$)
yes, and Age > 62.5: predict **COPD** (3 instances)
no (Smoking $= 2$): predict **Asthma** (7 instances)

On the six test records this tree predicts HC, HC, Asthma, HC, COPD, Asthma. None of those predictions matches the true label, so the test accuracy is $0/6 = 0$. The multi-way gain ratio of Smoking on the training set is 0.179595. The working is below.

3(a)(i) Induction

Training distribution

The training set has 18 instances. Class counts: HC 7, Infected 3, Asthma 5, COPD 3.

Root

Records 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18 ($n = 18$). Class counts: HC 7, Infected 3, Asthma 5, COPD 3.

$$H = -\left(\frac{7}{18}\log_2\frac{7}{18} + \frac{3}{18}\log_2\frac{3}{18} + \frac{5}{18}\log_2\frac{5}{18} + \frac{3}{18}\log_2\frac{3}{18}\right) = 1.904875.$$

Every binary split is evaluated below. For a numeric attribute the threshold is the midpoint between successive distinct values, which puts every instance of one value entirely on one side. For nominal Smoking, each binary split is one partition of the values that actually occur in the node.

Split	$\lvert L\rvert$	L counts	$H(L)$	$\lvert R\rvert$	R counts	$H(R)$	Gain
Smoking ∈ 1,3 vs Smoking ∈ 2	11	6, 2, 1, 2	1.685816	7	1, 1, 4, 1	1.664498	**0.227350**
Age ≤ 59.5 vs Age > 59.5	3	2, 1, 0, 0	0.918296	15	5, 2, 5, 3	1.908613	0.161315
Smoking ∈ 1,2 vs Smoking ∈ 3	14	5, 2, 5, 2	1.863121	4	2, 1, 0, 1	1.500000	0.122448
Age ≤ 62.5 vs Age > 62.5	12	6, 2, 3, 1	1.729574	6	1, 1, 2, 2	1.918296	0.112394
Age ≤ 58.5 vs Age > 58.5	1	1, 0, 0, 0	0.000000	17	6, 3, 5, 3	1.932805	0.079448
Smoking ∈ 1 vs Smoking ∈ 2,3	7	4, 1, 1, 1	1.664498	11	3, 2, 4, 2	1.936260	0.074300
Age ≤ 61.5 vs Age > 61.5	9	4, 1, 3, 1	1.752715	9	3, 2, 2, 2	1.974938	0.041049
Age ≤ 60.5 vs Age > 60.5	6	3, 1, 1, 1	1.792481	12	4, 2, 4, 2	1.918296	0.028517

Counts are ordered (HC, Infected, Asthma, COPD). The largest gain is in bold.

The best gain is 0.227350, which is at least 0.15, and the node has at least 8 instances, so the node is split on Smoking ∈ 1,3 vs Smoking ∈ 2.

$$\text{Gain} = H - \frac{11}{18}H(L) - \frac{7}{18}H(R) = 0.227350.$$

This gain is computed from the unrounded entropies and is the value compared with 0.15. The entropies in the table are those same quantities rounded to 6 decimal places.

Left child: Smoking $\in \{1, 3\}$

Records 1, 2, 3, 4, 7, 10, 11, 12, 13, 17, 18 ($n = 11$). Class counts: HC 6, Infected 2, Asthma 1, COPD 2.

$$H = -\left(\frac{6}{11}\log_2\frac{6}{11} + \frac{2}{11}\log_2\frac{2}{11} + \frac{1}{11}\log_2\frac{1}{11} + \frac{2}{11}\log_2\frac{2}{11}\right) = 1.685816.$$

Every binary split is evaluated below. For a numeric attribute the threshold is the midpoint between successive distinct values, which puts every instance of one value entirely on one side. For nominal Smoking, each binary split is one partition of the values that actually occur in the node.

Split	$\lvert L\rvert$	L counts	$H(L)$	$\lvert R\rvert$	R counts	$H(R)$	Gain
Age ≤ 62.5 vs Age > 62.5	8	5, 2, 1, 0	1.298795	3	1, 0, 0, 2	0.918296	**0.490793**
Age ≤ 61.5 vs Age > 61.5	5	3, 1, 1, 0	1.370951	6	3, 1, 0, 2	1.459148	0.266757
Age ≤ 60.5 vs Age > 60.5	4	2, 1, 1, 0	1.500000	7	4, 1, 0, 2	1.378783	0.262953
Age ≤ 59.5 vs Age > 59.5	3	2, 1, 0, 0	0.918296	8	4, 1, 1, 2	1.750000	0.162644
Age ≤ 58.5 vs Age > 58.5	1	1, 0, 0, 0	0.000000	10	5, 2, 1, 2	1.760964	0.084939
Smoking ∈ 1 vs Smoking ∈ 3	7	4, 1, 1, 1	1.664498	4	2, 1, 0, 1	1.500000	0.081135

Counts are ordered (HC, Infected, Asthma, COPD). The largest gain is in bold.

The best gain is 0.490793, which is at least 0.15, and the node has at least 8 instances, so the node is split on Age ≤ 62.5 vs Age > 62.5.

$$\text{Gain} = H - \frac{8}{11}H(L) - \frac{3}{11}H(R) = 0.490793.$$

This gain is computed from the unrounded entropies and is the value compared with 0.15. The entropies in the table are those same quantities rounded to 6 decimal places.

Left-left child: Smoking $\in \{1, 3\}$ and Age ≤ 62.5

Records 1, 2, 3, 4, 7, 10, 11, 12 ($n = 8$). Class counts: HC 5, Infected 2, Asthma 1.

$$H = -\left(\frac{5}{8}\log_2\frac{5}{8} + \frac{2}{8}\log_2\frac{2}{8} + \frac{1}{8}\log_2\frac{1}{8}\right) = 1.298795.$$

Every binary split is evaluated below. For a numeric attribute the threshold is the midpoint between successive distinct values, which puts every instance of one value entirely on one side. For nominal Smoking, each binary split is one partition of the values that actually occur in the node.

Split	$\|L\|$	L counts	$H(L)$	$\|R\|$	R counts	$H(R)$	Gain
Age ≤ 60.5 vs Age > 60.5	4	2, 1, 1, 0	1.500000	4	3, 1, 0, 0	0.811278	**0.143156**
Smoking $\in 1$ vs Smoking $\in 3$	6	4, 1, 1, 0	1.251629	2	1, 1, 0, 0	1.000000	0.110073
Age ≤ 59.5 vs Age > 59.5	3	2, 1, 0, 0	0.918296	5	3, 1, 1, 0	1.370951	0.097590
Age ≤ 61.5 vs Age > 61.5	5	3, 1, 1, 0	1.370951	3	2, 1, 0, 0	0.918296	0.097590
Age ≤ 58.5 vs Age > 58.5	1	1, 0, 0, 0	0.000000	7	4, 2, 1, 0	1.378783	0.092359

Counts are ordered (HC, Infected, Asthma, COPD). The largest gain is in bold.

The best gain is 0.143156, which is less than 0.15. Pre-pruning stops the split. The leaf label is the majority class HC.

Left-right child: Smoking $\in \{1, 3\}$ and Age > 62.5

Records 13, 17, 18 ($n = 3$). Class counts: HC 1, COPD 2.

$$H = -\left(\frac{1}{3}\log_2\frac{1}{3} + \frac{2}{3}\log_2\frac{2}{3}\right) = 0.918296.$$

This node has $n = 3 < 8$, so it is not split. The leaf label is the majority class COPD.

Right child: Smoking $= 2$

Records 5, 6, 8, 9, 14, 15, 16 ($n = 7$). Class counts: HC 1, Infected 1, Asthma 4, COPD 1.

$$H = -\left(\frac{1}{7}\log_2\frac{1}{7} + \frac{1}{7}\log_2\frac{1}{7} + \frac{4}{7}\log_2\frac{4}{7} + \frac{1}{7}\log_2\frac{1}{7}\right) = 1.664498.$$

This node has $n = 7 < 8$, so it is not split. The leaf label is the majority class Asthma.

3(a)(ii) Test evaluation

Record	Age	Smoking	Actual	Path	Predicted
19	58	1	Infected	$\{1,3\}$, then Age ≤ 62.5	HC
20	58	3	Asthma	$\{1,3\}$, then Age ≤ 62.5	HC
21	60	2	COPD	Smoking $= 2$	Asthma
22	62	3	COPD	$\{1,3\}$, then Age ≤ 62.5	HC
23	63	1	Asthma	$\{1,3\}$, then Age > 62.5	COPD
24	63	2	HC	Smoking $= 2$	Asthma

Record 22 has Age 62, and $62 \leq 62.5$, so it follows the HC leaf rather than the COPD leaf. With rows as the actual class and columns as the predicted class, in the order HC, Infected, Asthma, COPD:

Actual \ Predicted	HC	Infected	Asthma	COPD
HC	0	0	1	0
Infected	1	0	0	0
Asthma	1	0	0	1
COPD	1	0	1	0

No diagonal entry is positive, so

$$\text{Accuracy} = \frac{0}{6} = 0.$$

Gain ratio of the multi-way Smoking split

This calculation is separate from the tree. It uses one branch for each value of Smoking on the full training set. The parent entropy is the root entropy above, $H(D) = 1.904875$. Smoking $= 1$ ($n = 7$): HC 4, Infected 1, Asthma 1, COPD 1.

$$H = -\left(\frac{4}{7}\log_2\frac{4}{7} + \frac{1}{7}\log_2\frac{1}{7} + \frac{1}{7}\log_2\frac{1}{7} + \frac{1}{7}\log_2\frac{1}{7}\right) = 1.664498.$$

Smoking $= 2$ ($n = 7$): HC 1, Infected 1, Asthma 4, COPD 1.

$$H = -\left(\frac{1}{7}\log_2\frac{1}{7} + \frac{1}{7}\log_2\frac{1}{7} + \frac{4}{7}\log_2\frac{4}{7} + \frac{1}{7}\log_2\frac{1}{7}\right) = 1.664498.$$

Smoking $= 3$ ($n = 4$): HC 2, Infected 1, COPD 1.

$$H = -\left(\frac{2}{4}\log_2\frac{2}{4} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{4}\log_2\frac{1}{4}\right) = 1.500000.$$

$$\text{Remainder} = \frac{7}{18}\times 1.664498 + \frac{7}{18}\times 1.664498 + \frac{4}{18}\times 1.500000 = 1.627943.$$

$$\text{Gain} = 1.904875 - 1.627943 = 0.276932.$$

$$\text{SplitInfo} = -\left(\frac{7}{18}\log_2\frac{7}{18} + \frac{7}{18}\log_2\frac{7}{18} + \frac{4}{18}\log_2\frac{4}{18}\right) = 1.541982.$$

$$\text{GainRatio} = \frac{0.276932}{1.541982} = 0.179595.$$

Smoking $= 3$ has counts $(2, 1, 1)$ on four instances, so its entropy is exactly

$$-(\tfrac{1}{2}\log_2\tfrac{1}{2} + \tfrac{1}{4}\log_2\tfrac{1}{4} + \tfrac{1}{4}\log_2\tfrac{1}{4}) = \tfrac{1}{2} + \tfrac{1}{2} = 1.5.$$

Smoking $= 1$ and Smoking $= 2$ have the same count multiset $\{4, 1, 1, 1\}$, which is why their entropies agree.

Answer. GainRatio(Smoking) $= 0.179595$.

4 Question 4: Classification in Weka

The source file has 1,197 instances. The ARFF relation `productivity` keeps `team`, `smv`, `over_time`, `incentive`, and `no_of_workers` as numeric attributes, using the original numeric tokens. The class

$$\texttt{productivity} = \begin{cases} 1 & \text{if } \texttt{actual_productivity} \geq \texttt{targeted_productivity}, \\ 0 & \text{otherwise} \end{cases}$$

is declared nominal, `{0,1}`. None of these six fields is missing. The label is 1 on 875 instances and 0 on 322. Weka 3.8.6 was run from the Explorer classify pane, with the class set to `productivity` and the test option set to 10-fold cross-validation. The result text below is that Explorer output.

4(a) ZeroR

Weka Explorer — ZeroR, 10-fold cross-validation

```
=== Run information ===

Scheme:       weka.classifiers.rules.ZeroR
Relation:     productivity
Instances:    1197
Attributes:   6
              team
              smv
              over_time
              incentive
              no_of_workers
              productivity
Test mode:    10-fold cross-validation
```

Figure 1: ZeroR run information. The scheme is `weka.classifiers.rules.ZeroR` and the test mode is 10-fold cross-validation.

Weka Explorer — ZeroR confusion matrix (stratified cross-validation)

```
=== Confusion Matrix ===

   a   b   <-- classified as
   0 322 |   a = 0
   0 875 |   b = 1
```

Figure 2: Confusion matrix of ZeroR under stratified 10-fold cross-validation. Class `a` is productivity 0 and class `b` is productivity 1.

ZeroR predicts the majority class, which is 1. All 1,197 instances are classified as `b`. The 322 instances of class 0 are errors, so the cross-validated accuracy is $875/1197 = 73.0994\%$.

4(b) J48 with $C = 0.1$

Weka Explorer — J48, confidence factor C = 0.1

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.1 -M 2
Relation:     productivity
Instances:    1197
Attributes:   6
              team
              smv
              over_time
              incentive
              no_of_workers
              productivity
Test mode:    10-fold cross-validation
```

Figure 3: J48 run information. The confidence factor is $C = 0.1$ (`-C 0.1`); the default minimum leaf size $M = 2$ is unchanged. Test mode is 10-fold cross-validation.

Weka Explorer — J48 pruned tree (first lines)

```
J48 pruned tree
------------------

smv <= 3.9
|   no_of_workers <= 8: 0 (101.0/19.0)
|   no_of_workers > 8: 1 (27.0/7.0)
smv > 3.9
|   incentive <= 21
|   |   smv <= 4.6
|   |   |   no_of_workers <= 9
|   |   |   |   over_time <= 3000: 1 (177.0/56.0)
|   |   |   |   over_time > 3000: 0 (20.0/5.0)
|   |   |   no_of_workers > 9: 1 (150.0/28.0)
|   |   smv > 4.6: 0 (135.0/50.0)
|   incentive > 21
|   |   smv <= 29.4: 1 (512.0/27.0)
```

Figure 4: Opening of the J48 pruned tree trained on the full data. The figure contains the tree heading and more than the first ten lines of the tree.

4(c) CVParameterSelection on C

`CVParameterSelection` wraps J48. Its own cross-validation uses 10 folds, and the outer evaluation is also 10-fold cross-validation. The confidence factor C is searched at 16 values from 0.05 to 0.5 inclusive. Weka spaces those values by $(0.5 - 0.05)/(16 - 1) = 0.03$, namely

$$0.05, 0.08, 0.11, 0.14, \ldots, 0.47, 0.50.$$

The initial J48 options after `-` are the defaults (`-C 0.25 -M 2`); C is then replaced by the searched value. In the run-information figure the long scheme line is wrapped after `-S 1`. No tokens were removed.

Weka Explorer — CVParameterSelection + J48

```
=== Run information ===

Scheme:       weka.classifiers.meta.CVParameterSelection -P "C 0.05 0.5 16.0" -X 10 -S 1
              -W weka.classifiers.trees.J48 -- -C 0.25 -M 2
Relation:     productivity
Instances:    1197
Attributes:   6
              team
              smv
              over_time
              incentive
              no_of_workers
              productivity
Test mode:    10-fold cross-validation
```

Figure 5: Run information for CVParameterSelection. The parameter string is `C 0.05 0.5 16.0`, the internal fold count is `-X 10`, and the test mode is 10-fold cross-validation.

Weka Explorer — C selected by CVParameterSelection

```
Cross-validated Parameter selection.
Classifier: weka.classifiers.trees.J48
Cross-validation Parameter: '-C' ranged from 0.05 to 0.5 with 16.0 steps
Classifier Options: -C 0.14 -M 2
```

Figure 6: Parameter chosen on the full training set. The line `Classifier Options: -C 0.14 -M 2` is the selected setting.

Selected value. The confidence factor selected by CVParameterSelection is $C = 0.14$.

5 Question 5: Classification in Python

The program is `src/q5_abalone.py`. It reads `abalone.data` with no header. The columns in file order are Sex, Length, Diameter, Height, Whole weight, Shucked weight, Viscera weight, Shell weight, and Rings. The new table keeps Length, Diameter, Height, and Whole weight, and sets

$$\text{Mature} = \begin{cases} 1 & \text{if Rings} > 8, \\ 0 & \text{otherwise.} \end{cases}$$

The first three raw ring counts are 15, 7, and 9, so the first three labels are 1, 0, and 1. The 8:2 split is `train_test_split` with `test_size=0.2` and `random_state=42` (3,341 training rows and 836 test rows; $3341 + 836 = 4177$). Grid search uses only the training rows, `DecisionTreeClassifier`, scoring `f1` (binary F1 for the positive label 1), and the default 5-fold cross-validation. The grid is `criterion` $\in \{\text{gini}, \text{entropy}\}$ and `max_depth` $\in \{3, \ldots, 10\}$.

5(a) Step 1

```
from pathlib import Path

import pandas as pd
from sklearn.model_selection import GridSearchCV, train_test_split
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

DATA_PATH = Path(__file__).resolve().parents[1] / "data" / "abalone.data"

# ---------------------------------------------------------------------
# Step 1. Read abalone.data and construct the new dataset.
# ---------------------------------------------------------------------
columns = [
    "Sex",
    "Length",
    "Diameter",
    "Height",
    "Whole weight",
    "Shucked weight",
    "Viscera weight",
    "Shell weight",
    "Rings",
]
raw = pd.read_csv(DATA_PATH, header=None, names=columns)
data = pd.DataFrame(
    {
        "Length": raw["Length"],
        "Diameter": raw["Diameter"],
        "Height": raw["Height"],
        "Whole weight": raw["Whole weight"],
        "Mature": (raw["Rings"] > 8).astype(int),
    }
)

print("===== Step 1 output =====")
print("Preview of the first 3 rows of the new dataset:")
print(data.head(3).to_string(index=True))
print(f"Number of rows: {len(data)}")
print(f"Number of columns: {data.shape[1]}")
```

python3 src/q5_abalone.py — Step 1

```
===== Step 1 output =====
Preview of the first 3 rows of the new dataset:
   Length  Diameter  Height  Whole weight  Mature
0   0.455     0.365   0.095        0.5140       1
1   0.350     0.265   0.090        0.2255       0
2   0.530     0.420   0.135        0.6770       1
Number of rows: 4177
Number of columns: 5
```

Figure 7: Step 1 output: the first three rows of the new dataset, and its shape 4177×5.

5(b) Steps 2 and 3

```python
# Step 2. Split into training and testing data, ratio 8:2.
# ---------------------------------------------------------------------------
features = ["Length", "Diameter", "Height", "Whole weight"]
X = data[features]
y = data["Mature"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ---------------------------------------------------------------------------
# Step 3. Tune DecisionTreeClassifier with GridSearchCV on training data.
#         Score is the F1-score of the positive class (Mature = 1).
# ---------------------------------------------------------------------------
param_grid = {
    "criterion": ["gini", "entropy"],
    "max_depth": [3, 4, 5, 6, 7, 8, 9, 10],
}
search = GridSearchCV(
    estimator=DecisionTreeClassifier(random_state=42),
    param_grid=param_grid,
    scoring="f1",
    cv=5,
)
search.fit(X_train, y_train)

print("\n===== Step 2 and Step 3 output =====")
print(f"Training set size: {len(X_train)}")
print(f"Testing set size: {len(X_test)}")
print(f"Split ratio (train:test) = {len(X_train)}:{len(X_test)}")
print(f"Best parameters: {search.best_params_}")
print(f"Best cross-validated F1-score (Mature = 1): {search.best_score_:.4f}")
```

python3 src/q5_abalone.py — Steps 2 and 3

```
===== Step 2 and Step 3 output =====
Training set size: 3341
Testing set size: 836
Split ratio (train:test) = 3341:836
Best parameters: {'criterion': 'gini', 'max_depth': 4}
Best cross-validated F1-score (Mature = 1): 0.8729
```

Figure 8: Steps 2 and 3 output. The selected parameters are `criterion=gini` and `max_depth=4`. The best cross-validated F1-score for Mature = 1 is 0.8729.

5(c) Step 4

```
82 # Step 4. Fit a decision tree with the selected parameters and print it.
83 # ------------------------------------------------------------------------
84 best_tree = DecisionTreeClassifier(random_state=42, **search.best_params_)
85 best_tree.fit(X_train, y_train)
86 tree_text = export_text(best_tree, feature_names=features, decimals=4)
87
88 print("\n===== Step 4 output =====")
89 print(f"Decision tree parameters: {search.best_params_}")
90 print("Decision tree (text):")
91 print(tree_text)
```

python3 src/q5_abalone.py — Step 4 (text tree, part 1)

```
===== Step 4 output =====
Decision tree parameters: {'criterion': 'gini', 'max_depth': 4}
Decision tree (text):
|--- Height <= 0.1225
|   |--- Height <= 0.1025
|   |   |--- Whole weight <= 0.1517
|   |   |   |--- Whole weight <= 0.1237
|   |   |   |   |--- class: 0
|   |   |   |--- Whole weight >  0.1237
|   |   |   |   |--- class: 0
|   |   |--- Whole weight >  0.1517
|   |   |   |--- Diameter <= 0.3825
|   |   |   |   |--- class: 0
|   |   |   |--- Diameter >  0.3825
|   |   |   |   |--- class: 1
|   |--- Height >  0.1025
|   |   |--- Whole weight <= 0.6090
|   |   |   |--- Whole weight <= 0.5948
|   |   |   |   |--- class: 0
|   |   |   |--- Whole weight >  0.5948
|   |   |   |   |--- class: 0
|   |   |--- Whole weight >  0.6090
|   |   |   |--- Diameter <= 0.3825
|   |   |   |   |--- class: 0
|   |   |   |--- Diameter >  0.3825
|   |   |   |   |--- class: 1
|--- Height >  0.1225
|   |--- Height <= 0.1525
```

Figure 9: Step 4 output, first part of the text tree. The fitted parameters are gini and max depth 4.

python3 src/q5_abalone.py — Step 4 (text tree, part 2)

```
|   |   |--- Whole weight <= 0.7945
|   |   |   |--- Diameter <= 0.2925
|   |   |   |   |--- class: 0
|   |   |   |--- Diameter >  0.2925
|   |   |   |   |--- class: 1
|   |   |--- Whole weight >  0.7945
|   |   |   |--- Diameter <= 0.3775
|   |   |   |   |--- class: 0
|   |   |   |--- Diameter >  0.3775
|   |   |   |   |--- class: 1
|   |--- Height >  0.1525
|   |   |--- Length <= 0.4975
|   |   |   |--- Diameter <= 0.4175
|   |   |   |   |--- class: 1
|   |   |   |--- Diameter >  0.4175
|   |   |   |   |--- class: 0
|   |   |--- Length >  0.4975
|   |   |   |--- Diameter <= 0.4825
|   |   |   |   |--- class: 1
|   |   |   |--- Diameter >  0.4825
|   |   |   |   |--- class: 1
```

Figure 10: Step 4 output, remainder of the same text tree.

5(d) Step 5

```
# Step 5. Evaluate on the testing data for class Mature = 1.
# ---------------------------------------------------------------------
y_pred = best_tree.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, pos_label=1)
recall = recall_score(y_test, y_pred, pos_label=1)
f1 = f1_score(y_test, y_pred, pos_label=1)

print("===== Step 5 output =====")
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision (Mature = 1): {precision:.4f}")
print(f"Recall (Mature = 1): {recall:.4f}")
print(f"F1-score (Mature = 1): {f1:.4f}")
```

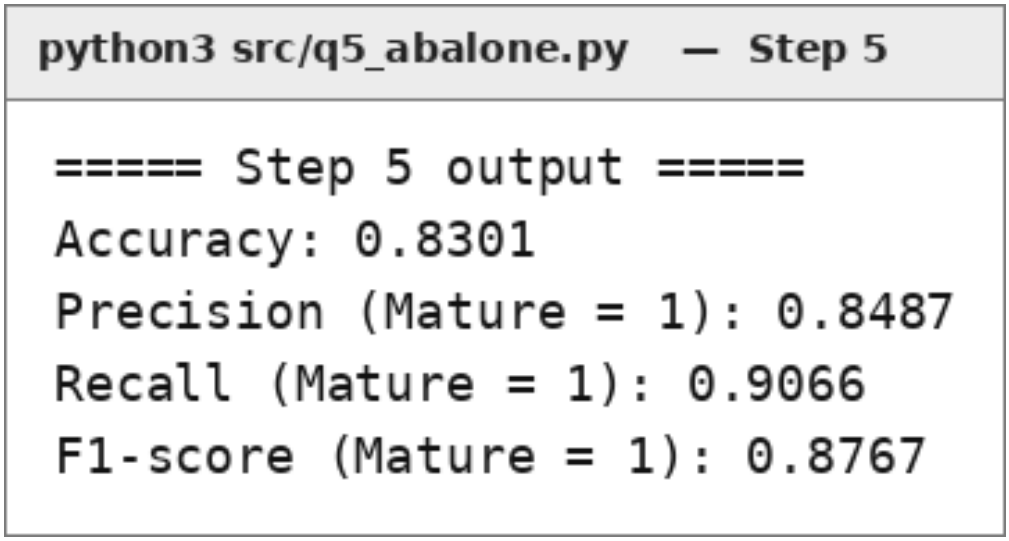


Figure 11: Step 5 output on the held-out 20%, scored for class Mature = 1.

On the test set the decision tree has accuracy 0.8301, precision 0.8487, recall 0.9066, and F1-score 0.8767, all for Mature = 1 except accuracy, which is the overall accuracy.